# Category Theory for Mathematical Physics — Exercise Sheet 11

*Algebras in tensor categories and module categories*

**Variant: Questions only**

Conventions as in Lecture 11: $C$ is a fusion category, $\mathcal{M}$ a left $C$-module category, $[M,N]$ the internal Hom (Definition 11.7); $\psi$ is a 2-cochain (in Exercise 6, $\psi$ is Ising's fermion instead), and $(\delta\psi)$, $\omega$ are as in Lecture 3. The *NIM-rep* of a semisimple module category $\mathcal{M}$ with simple objects $M_1,\dots,M_r$ is the family of matrices $(n_a)_{ij} = \dim \mathrm{Hom}_{\mathcal{M}}(a \otimes M_j, M_i)$, one for each simple $a$ of $C$. In general, a *NIM-rep* of the fusion ring of $C$ is a family of square non-negative integer matrices $n_a$, all of one size $r \geq 1$, with $n_1 = 1$ for the unit ($n_0$ in Exercise 5), $n_a n_b = \sum_c N_{ab}^c\, n_c$ and $n_{a^*} = n_a^T$; it is *indecomposable* if no nonempty proper set of indices (simple objects) is closed under every $n_a$. Module categories on this sheet are nonzero. $\mathbb{Z}_N$ is additive, $\mathbb{Z}_2 = \{1, s\}$ and $S_3$ multiplicative. In Exercise 2, $\varepsilon$ is Sheet 3's carry (not the toric fermion) and $\zeta_4 = i$. In Exercise 5, $U_m$ is the polynomial with $U_0 = 1$, $U_1 = x$, $U_{m+1} = xU_m - U_{m-1}$, so that $U_m(2\cos y) = \sin((m+1)y)/\sin y$; in Exercise 6, $\pi$ is an involution of the simple objects.

**Exercise 1 (Module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$).** (i) Using Theorem 11.11, list the indecomposable semisimple module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$, with their simple objects and the matrix $n_s$.

(ii) The direct sum $\mathrm{Vec}_{\mathbb{Z}_2} \oplus \mathrm{Vect}_{\mathrm{fd}}$ is a module category too. Write its $n_s$, and say why it is not on the list.

**Exercise 2 (The algebra of a rank-one module).** Let $\omega_p$ be the cocycle of Sheet 3 on $\mathbb{Z}_4$ and $H = \{0, 2\}$.

(i) Write the module pentagon of Theorem 11.6 at $(g,h,k) = (2,2,2)$ for a normalised $\psi$, and show that a rank-one module category over the part of $\mathrm{Vec}_{\mathbb{Z}_4}^{\omega_p}$ on $H$ exists exactly when $p$ is even.

(ii) For such a module $\mathcal{M} = \mathrm{Vect}_{\mathrm{fd}}$, with $M = \mathbb{C}$, compute $[M,M]$ by Theorem 11.8 and its product on the basis $e_h \in \mathrm{Hom}(\delta_h \otimes M, M)$, and show that it is the twisted group algebra $\mathbb{C}_\psi[H]$ of Example 11.2.

(iii) For $\mathrm{Mod}(\mathbb{C}[H])$ over $\mathrm{Vec}_G$ (trivial $\omega$, $H \leq G$), show that $[M_H, M_H] \cong \mathbb{C}[H]$ as algebras.

**Exercise 3 (The four module categories over $\mathrm{Vec}_{S_3}$).** You may use that $H^2(H, \mathbb{C}^\times) = 0$ for every subgroup $H$ of $S_3$.

(i) List the subgroups of $S_3$ up to conjugacy, and the module categories of Theorem 11.11, with their numbers of simple objects.

(ii) For $H = \mathbb{Z}_3$, write the action of a transposition and of a 3-cycle on the simple objects.

(iii) Over $\mathrm{Rep}(S_3)$ the same pairs give $\mathrm{Rep}(H)$ (Lecture 11, cited). How many simple objects does each have?

**Exercise 4 (NIM-reps; Fibonacci).** (i) Show that the NIM-rep of a semisimple module category satisfies $n_1 = 1$ and $n_a n_b = \sum_c N_{ab}^c\, n_c$; you may use that $n_{a^*} = n_a^T$ (duality).

(ii) For Fibonacci, show that a symmetric matrix $n$ of size $r \geq 1$ with non-negative integer entries and $n^2 = 1 + n$ that is indecomposable (no nonempty proper set of simples closed under $n$) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ up to relabelling. Which module category has this NIM-rep?

(iii) Let $\mathcal{M}$ be a nonzero indecomposable semisimple module category over Fibonacci, with finitely many simple objects. Show that its NIM-rep is indecomposable. Then take a simple $M$ with $(n_\tau)_{MM} = 0$, and use Theorems 11.8 and 11.10 to show that $\mathcal{M}$ is equivalent to the regular module $C$.

**Exercise 5 (**$\mathrm{SU}(2)_4$**:** $A_5$ **and** $D_4$**).** For $\mathrm{SU}(2)_4$, a NIM-rep is determined by $n = n_{1/2}$ through $n_0 = 1$ and $n_{j+1/2} = n\, n_j - n_{j-1/2}$ (Sheet 5, Exercise 2).

(i) The Dynkin diagram $A_5$ is the path on the vertices $1, 2, 3, 4, 5$, and $D_4$ is the graph with a centre 2 joined to three ends $1, 3, 4$. Draw them, and let $n$ be the adjacency matrix of either. Show that the $n_j$, $j = 0, \dots, 2$, are non-negative integer matrices and that $n_{5/2} = 0$, and deduce that the $n_j$ form a NIM-rep.

(ii) Show that the eigenvalues of $n$ are $2\cos(m\pi/6)$ for $m = 1, \dots, 5$ ($A_5$) and $m = 1, 3, 3, 5$ ($D_4$), and explain why $n_{5/2} = 0$ follows from this.

**Exercise 6 ⋆.** *Ising has only the regular module.* A NIM-rep of the Ising ring is a pair of symmetric $r \times r$ ($r \geq 1$) non-negative integer matrices $n_\sigma$, $n_\psi$ with $n_\psi^2 = 1$, $n_\sigma n_\psi = n_\sigma$ and $n_\sigma^2 = 1 + n_\psi$.

(i) Show that $n_\psi$ is the permutation matrix of an involution $\pi$ of the simple objects, and that each row of $n_\sigma$ is unchanged by $\pi$.

(ii) Show that if $\pi$ moves $i$, then row $i$ of $n_\sigma$ has a single 1, at a point $j$ that $\pi$ fixes, and row $j$ is $e_i + e_{\pi(i)}$.

(iii) Show that $\pi$ cannot fix every simple object, and conclude that every indecomposable NIM-rep is the regular one, with three simple objects.

(iv) Conclude, as in Exercise 4(iii), that every nonzero indecomposable semisimple module category over Ising, with finitely many simple objects, is equivalent to the regular one.